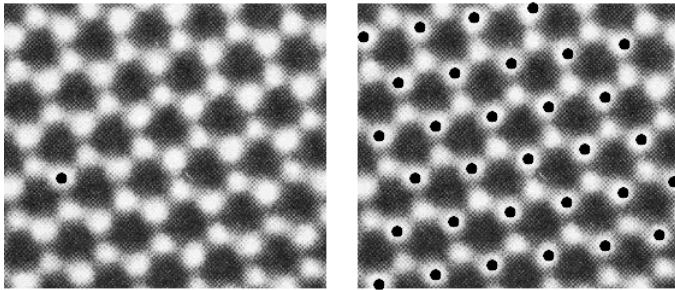


1.7.5: Lattices

Crystalline structures are characterised by a repeating pattern in three dimensions. The periodic nature of the structure can be represented using a lattice.

To generate the lattice from any repeating pattern, we choose an arbitrary reference point and examine its environment. We then simply mark in all the points in the pattern that are identical to the chosen reference point. The set of identical points is the lattice, and each point within it is a **lattice point**.



A. Putnis, *Introduction to Mineral Sciences*, Cambridge University Press, 1992

Note that not all white discs within this pattern are exactly equivalent, and therefore they are not all lattice points. The discs marked with a black spot have different arrangements around them than those that are unmarked (each is surrounded by 3 others in a triangle, but the orientation of the triangles is different).

To practice identifying the lattice points within a more complex repeating pattern, try the following game! The brick pattern corresponds to an unusual style of Danish bricklaying where after each normal brick, the next is laid breadthwise, and so on. The original image was taken from the wall of the Centre for Electron Nanoscopy in Copenhagen (thanks to Dr Rafal Dunin-Borkowski).

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